DP Biology – C1.3

Photosynthesis

- How does photosynthesis capture and use the energy in sunlight?
- How do non-living factors affect photosynthesis?

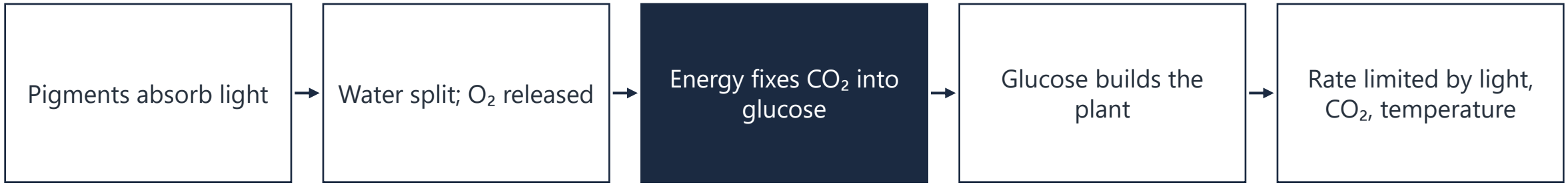
SL statements C1.3.1–8. Textbook pp. 425–438.
Image: light passing through a leaf. Jon Sullivan, public domain.



Objectives for each class

Class	Statements	By the end of the class, you can…
1	C1.3.1–3	Explain the energy change in photosynthesis, write its word equation, and explain where the oxygen comes from
2	C1.3.4–6	Separate pigments by chromatography, calculate Rf values, and compare absorption and action spectra
3	C1.3.7–8	Investigate limiting factors, interpret data, and evaluate CO_2 enrichment experiments

The big picture: capturing sunlight



Photosynthesis uses light energy to build carbon compounds from CO_2 and water.

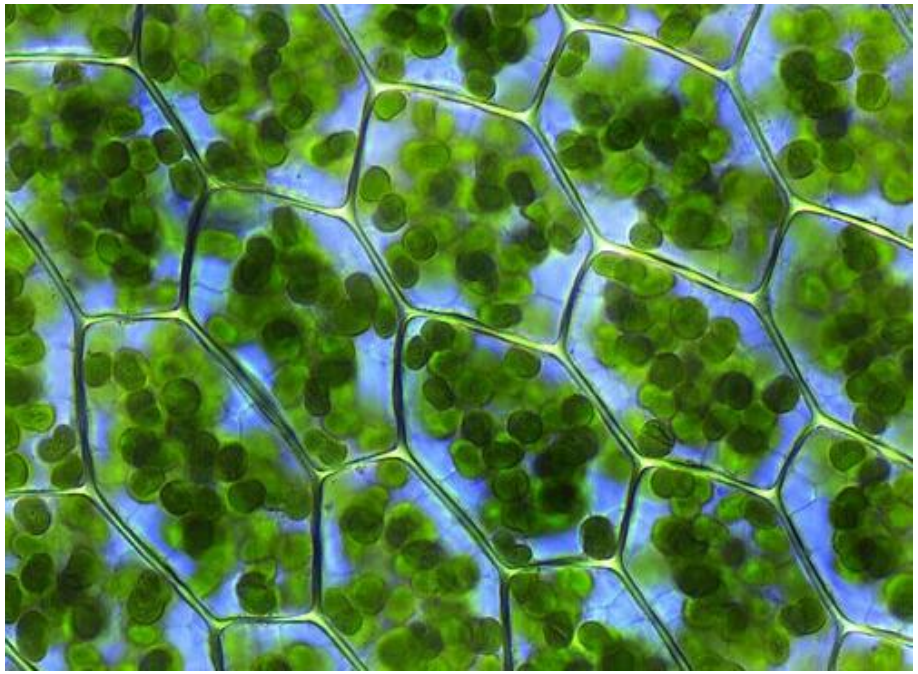
Those compounds feed the plant and, through food chains, almost every other organism.

Big question

A large oak tree has a mass of many tonnes. Where did all that mass come from?

From light energy to chemical energy

C1.3.1



Chloroplasts in moss cells. Kristian Peters, CC BY-SA 3.0.

Photosynthesis transforms **light energy** into **chemical energy** in carbon compounds such as glucose.

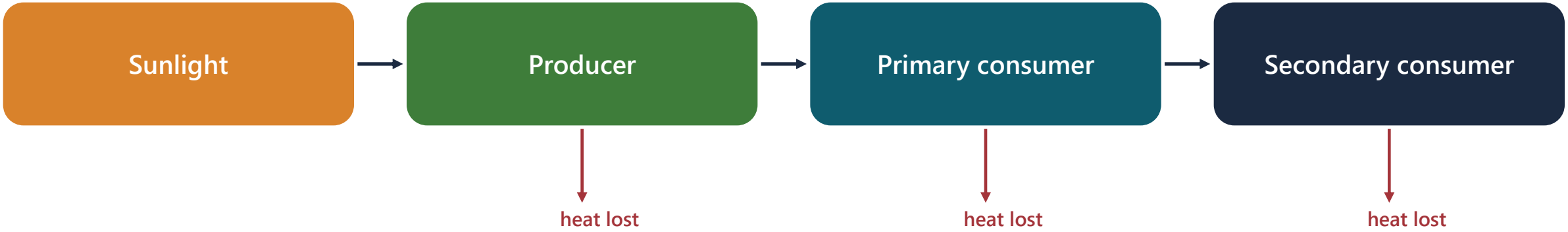
In plants and algae it happens in **chloroplasts**. Organisms that make their own carbon compounds are **autotrophs**.

Key idea

Energy is not made; it is transformed from light into chemical energy in bonds.

Photosynthesis powers ecosystems

C1.3.1



Producers make carbon compounds that **heterotrophs** eat. Energy is lost as heat at each step, so ecosystems need a constant supply of new energy from sunlight.

Link

C4.2: food chains, energy pyramids and the carbon cycle.

The word equation

C1.3.2

carbon dioxide + water → glucose + oxygen

(using light energy, absorbed by chlorophyll)



Big question

A tree's mass comes almost entirely from CO_2 in the air and water, not from the soil.

Oxygen is a by-product

C1.3.3



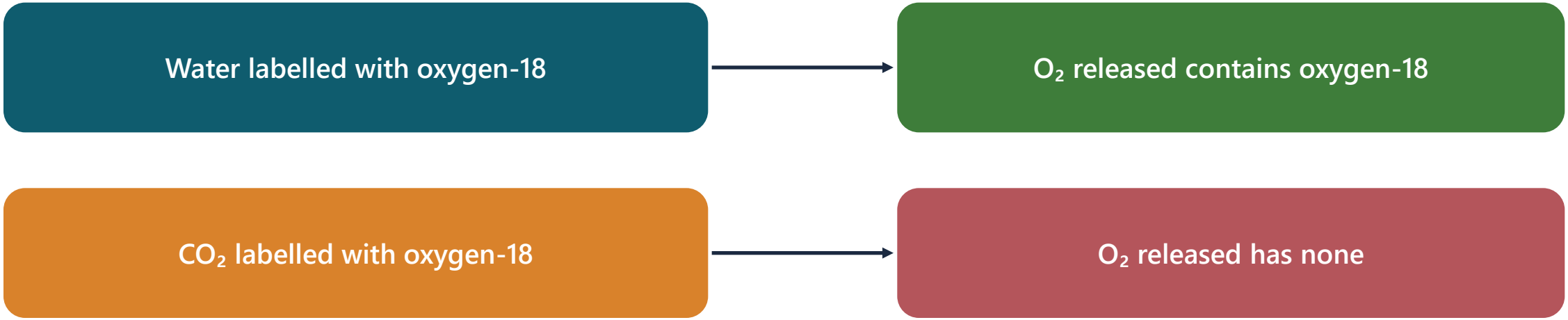
Stromatolites: living mats of cyanobacteria. Happy Little Nomad, CC BY-SA 2.0.

The oxygen released comes from **splitting water** (photolysis). Some is used in respiration; the rest diffuses out as a **by-product**.

Oxygen from early cyanobacteria built up over hundreds of millions of years, making aerobic life possible.

Evidence: the oxygen comes from water

C1.3.3



Ruben and Kamen (1941) traced a heavy isotope of oxygen. Only labelled **water** gave labelled oxygen gas.

Nature of science

Isotopes let scientists trace atoms through a reaction: new techniques bring new understanding.

Check-in: class 1

C1.3.1–3

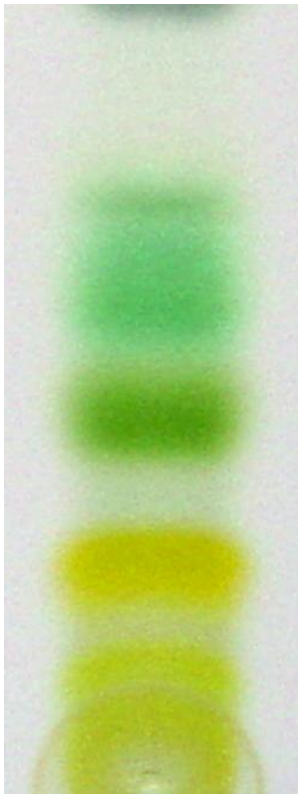
1. State the energy change in photosynthesis.
2. Write the word equation.
3. State the source of the oxygen released.
4. Describe the evidence that it comes from water.
5. Explain why animals depend on photosynthesis.

How to use this

Answer without your notes, then check against slides 3–8.

Separating leaf pigments by chromatography

C1.3.4



Pigments on a strip. Flo, CC BY-SA 2.5.

1. Grind a leaf with solvent to extract pigments.
2. Spot the extract on a **pencil line**.
3. Stand the strip in solvent, spot above the surface.
4. Pigments move different distances because they differ in solubility.
5. Mark the solvent front.

Pigment	Colour
Carotene	Orange-yellow
Xanthophyll	Yellow
Chlorophyll a	Blue-green
Chlorophyll b	Yellow-green

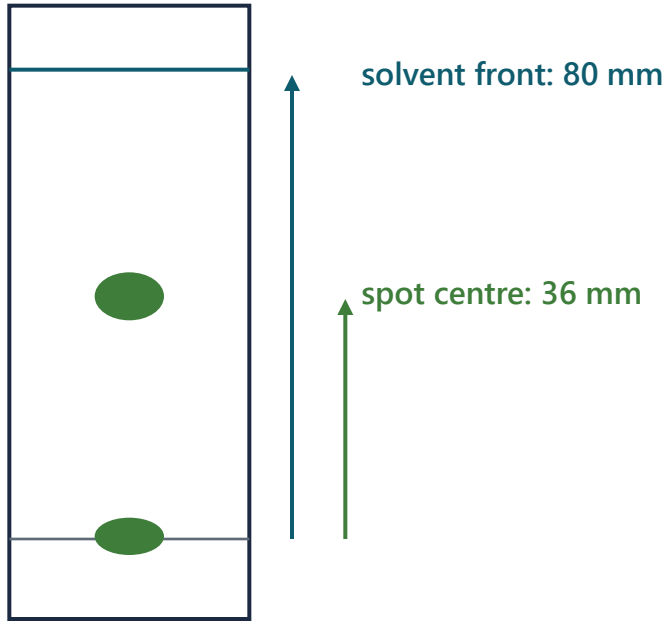
Safety
Ventilated room, no flames, eye protection.

Why pencil?
Ink would dissolve and separate too.

Identifying pigments with Rf values

Formula

Rf = distance moved by pigment / distance moved by solvent



Measure from the **pencil line** to the **centre** of the spot.

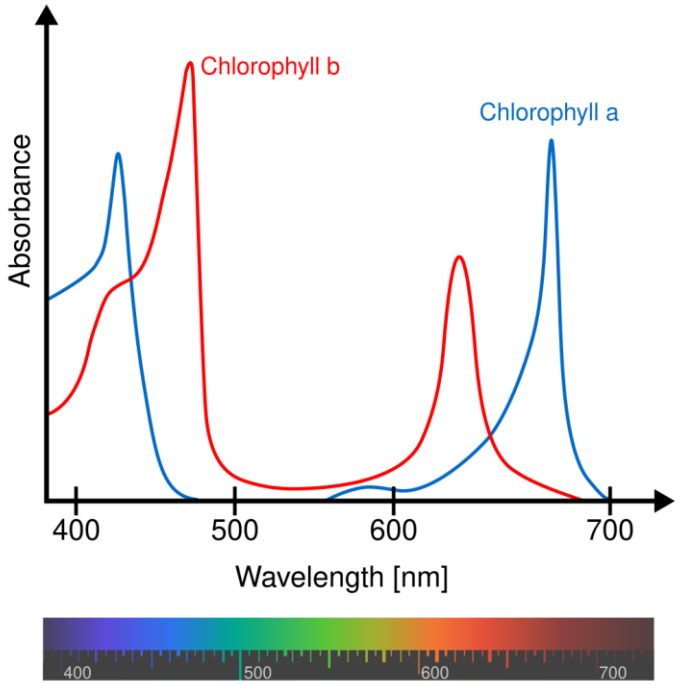
Rf is between 0 and 1 and has **no units**. Identify pigments by colour as well.

Worked example

Rf = 36 / 80 = 0.45 (chlorophyll b in this solvent).

How pigments absorb light

C1.3.5



Light comes as **photons**; each wavelength carries a different energy.

A pigment absorbs a photon and an **electron is excited**: light energy becomes chemical energy.

Chlorophyll absorbs **red** and **blue** and **reflects green**.

Absorption spectra of chlorophyll a and b. Daniele Pugliesi, CC BY-SA 3.0.

Exam tip

Label spectra with wavelengths (about 400 to 700 nm) and colours.

Why leaves have several pigments

C1.3.5

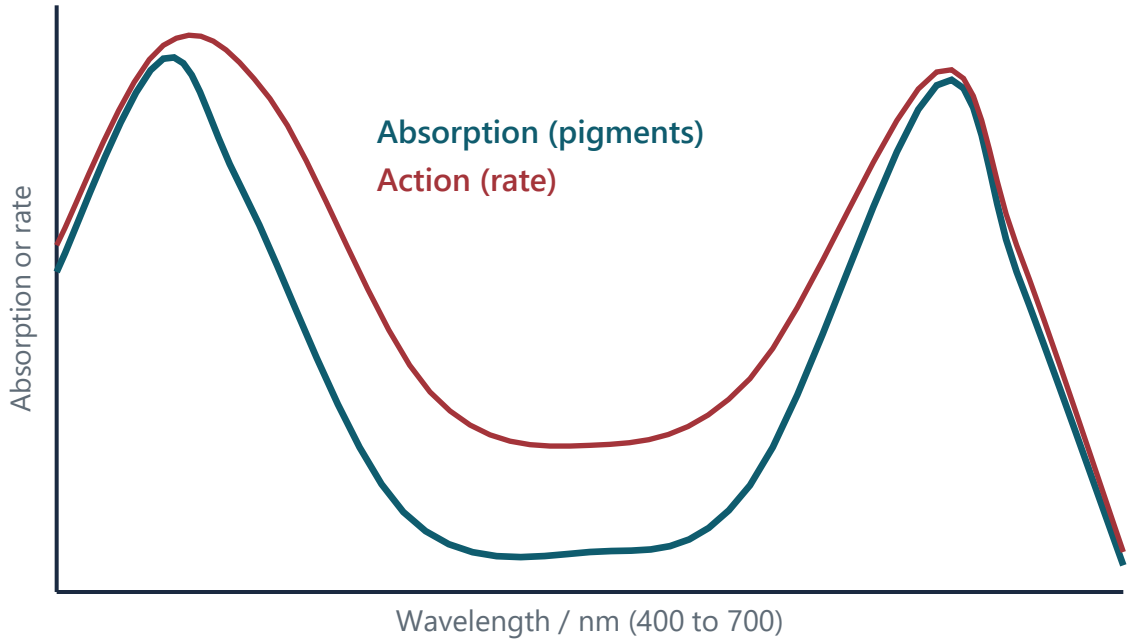
Pigment	Absorbs mainly	Role
Chlorophyll a	Blue and red	Main pigment
Chlorophyll b	Blue and orange-red	Widens the range used
Carotenoids	Blue and blue-green	Widen range; protect from excess light

Autumn colours

Trees break down chlorophyll in autumn. The yellow and orange carotenoids, there all along, become visible.

Absorption and action spectra

C1.3.6



General shapes only. Blue left, red right.

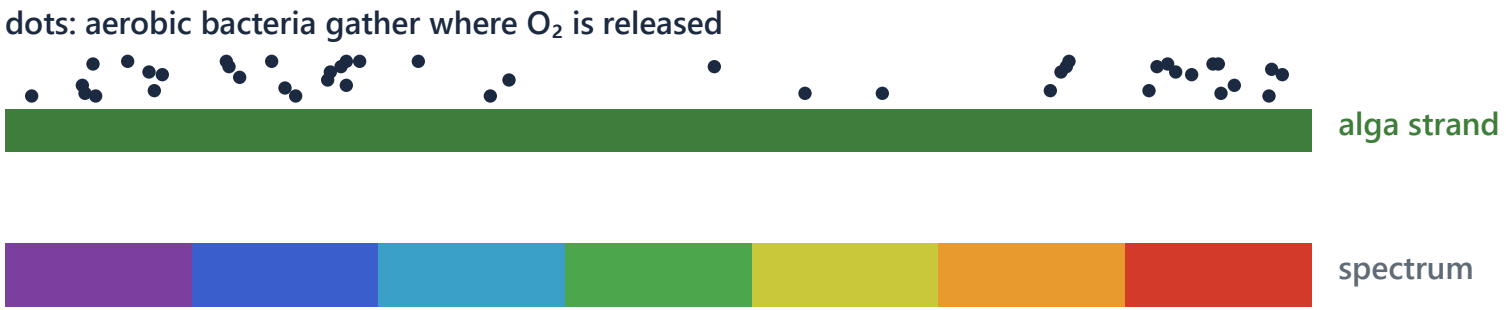
	Absorption	Action
y-axis	% absorbed	Rate
Peaks	Blue, red	Blue, red
Green	Little absorbed	Low, not zero

Why similar?

Light must be absorbed to drive photosynthesis, so the action spectrum follows absorption. Accessory pigments raise it in places.

Engelmann's experiment

C1.3.6



1882: Engelmann lit an alga with a spectrum. **Aerobic bacteria** gathered at **blue** and **red** light, few at green: a living action spectrum.

Discuss

Name the independent and dependent variables. What did he assume about the bacteria?

Practice: pigments and spectra

C1.3.4–6

Calculate

Spot 24 mm, solvent 60 mm. Rf?

Explain

Why does chlorophyll look green?

Predict

Rate in pure green light: zero or low?

Sketch

Draw an absorption and an action spectrum on the same axes and label the peaks.

Check-in: class 2

C1.3.4–6

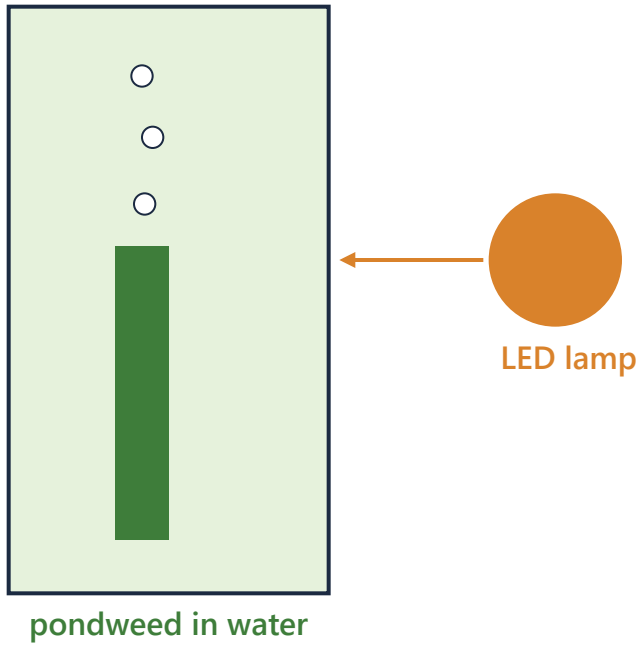
1. Explain why pigments separate in chromatography.
2. State why Rf has no units.
3. Describe what happens when a pigment absorbs light.
4. Distinguish absorption and action spectra.
5. Explain what Engelmann’s bacteria showed.

How to use this

Answer without your notes, then check against slides 10–16.

Measuring the rate of photosynthesis

C1.3.7

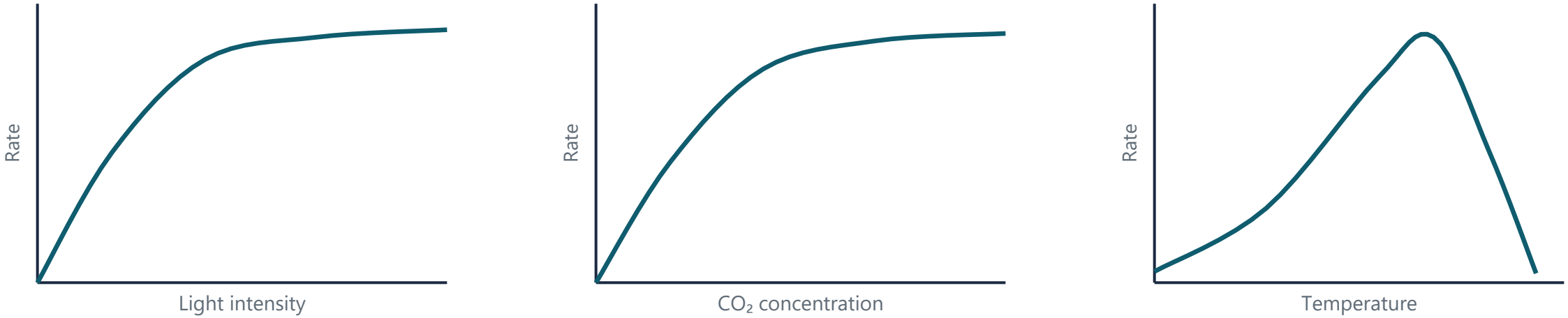


Measure	How
O_2 released	Count bubbles or collect gas
CO_2 taken in	pH indicator rises
Biomass	Dry mass before and after

Vary	How
Light	Lamp distance
CO_2	Sodium hydrogencarbonate
Temperature	Water bath

Limiting factors

C1.3.7



General shapes only.

The factor in shortest supply is the **limiting factor**: raising it raises the rate until another factor limits.

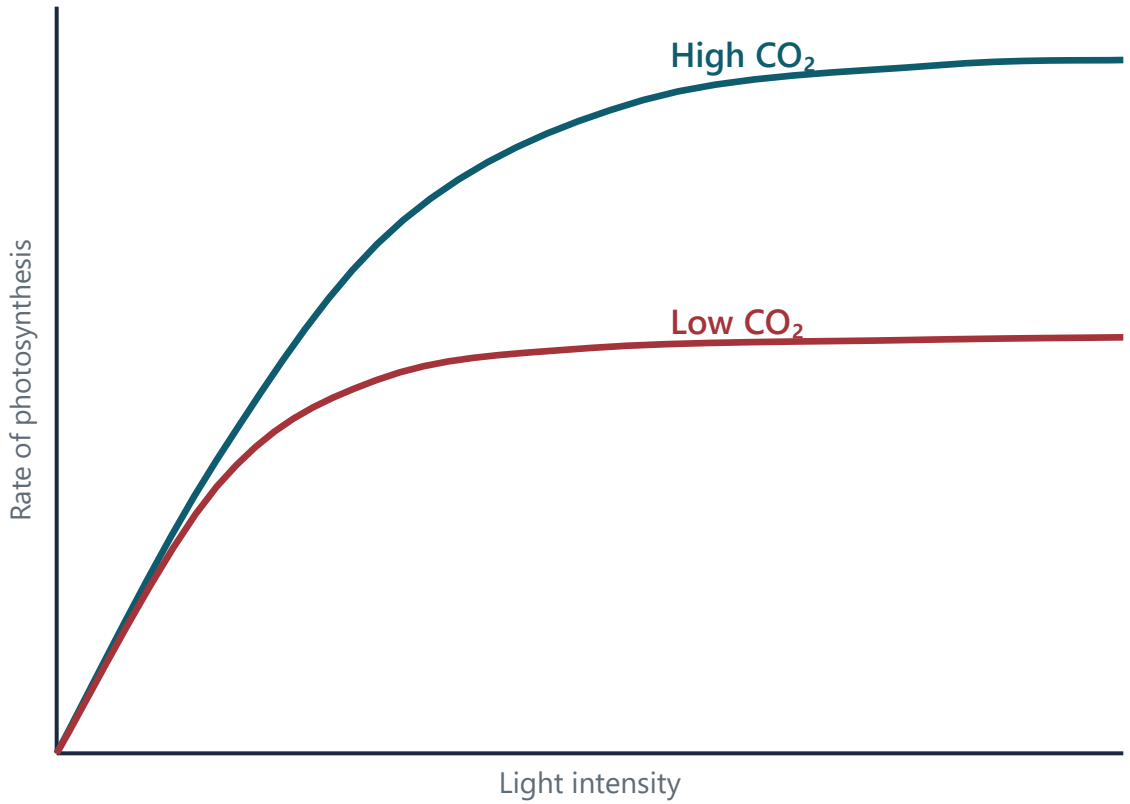
Temperature has an **optimum**: above it, enzymes denature.

Try it

On the light graph, what limits the rising part? What could limit the plateau?

Reading a graph with two limiting factors

C1.3.7



Low light: lines rise together; light limits.

High light: low-CO_2 line levels off first; CO_2 limits.

More CO_2 lifts the plateau until light or temperature limits.

Working with data: pondweed and light

C1.3.7

Bubbles per minute at different distances from an LED, 25 °C. Invented values; realistic pattern.

Distance / cm	Relative light	Bubbles / min
10	100	48
20	25	30
30	11	17
40	6	9
50	4	6

1. Describe the effect of distance.
2. Show how 25 at 20 cm was calculated.
3. Why not four times faster at 10 cm than 20 cm?
4. Suggest one improvement.

Carbon dioxide enrichment experiments

C1.3.8



A FACE facility in Nevada. NNSA, public domain.

CO_2 in the air is rising. Will plants grow faster? Enrichment experiments raise CO_2 and measure growth.

	Greenhouse	FACE (free-air)
Method	CO_2 added to a closed space	CO_2 released over open plots
Strength	Other factors controlled	Real soil, weather, species

Growth usually rises at first, by less than expected, then levels off as nitrogen or water limit.

Check-in: class 3

C1.3.7–8

1. Explain the term limiting factor.
2. Explain why rate falls above an optimum temperature.
3. Describe how to vary light intensity in an experiment.
4. Interpret a graph with high and low CO_2 lines.
5. Give one advantage of FACE over greenhouse experiments.

How to use this

Answer without your notes, then check against slides 18–22.

Common mistakes to avoid

Not this	But this
Plants get their mass from the soil.	Mostly from CO_2 (and water).
The oxygen comes from CO_2.	It comes from splitting water.
Chlorophyll absorbs green light.	It reflects green; absorbs red and blue.
Plants photosynthesize instead of respiring.	They do both; respiration all the time.
Rf has units of mm.	Rf is a ratio, with no units.

Summary: photosynthesis

C1.3.1–8

	Key idea
C1.3.1	Light to chemical energy; powers ecosystems
C1.3.2	CO_2 + water → glucose + oxygen
C1.3.3	O_2 from splitting water
C1.3.4	Chromatography and Rf
C1.3.5	Pigments absorb red and blue
C1.3.6	Absorption and action spectra
C1.3.7	Limiting factors
C1.3.8	CO_2 enrichment experiments

Check that you can...

1. **State** the energy change in photosynthesis
2. **Write** the word equation
3. **Explain** the source of the oxygen
4. **Calculate** Rf values
5. **Compare** absorption and action spectra
6. **Interpret** limiting-factor graphs and **evaluate** enrichment experiments

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Write the word equation for photosynthesis.

2. A pigment moves 24 mm, and the solvent moves 60 mm. Calculate Rf.

3. State where the oxygen released in photosynthesis comes from.

Image credits

Leaf 1 web.jpg – Jon Sullivan – Public domain

Plagiomnium affine laminazellen.jpeg – Kristian Peters – Fabelfroh – CC BY-SA 3.0

Stromatolites in Shark Bay.jpg – Happy Little Nomad – CC BY-SA 2.0

Chromatography of chlorophyll – Step 7.jpg – Flo~commonswiki – CC BY-SA 2.5

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